

Advancing AI Incidents Classification: Leveraging LLMs with Strategic Prompting

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Abstract. This study examines the efficacy of large language models (LLMs), particularly GPT-4, in classifying AI incident reports documented in the AI Incidents Database (AIID), with the goal of enhancing our understanding and management of AI-related harm. The data from incident reports are collected from news events detailing specific incidents related to AI technology that have resulted in harmful effects on our society. We explore the use of different prompting techniques on GPT-4 and assess the classification results of those incidents by subjective and objective evaluations. This work lays the groundwork for a comprehensive, automated classification framework for AI incident reporting, balancing LLM capabilities with the intricacies inherent in human judgment.¹

Keywords: Large Language Models · LLM Classification · Prompt Engineering · Responsible AI.

1 Introduction

The proliferation of artificial intelligence (AI) technologies has introduced unprecedented opportunities for innovation across diverse sectors. However, with these advancements come emerging concerns about AI failures. Instances of AI causing harm and sparking controversies underscore the critical need for better AI governance. To understand the impact of these technological setbacks, we use data visualization and analysis to outline the nature and scale of the harm. Hence, it is essential to develop an effective classification framework to systematically categorize the data for our study.

The primary source of data for our analysis and visualization comes from the AI Incidents Database (AIID)². The AIID serves as a repository of news articles documenting instances where AI technologies have caused harmful outcomes in the real-world. As part of a larger study of Responsible AI [1], this paper

¹ Our code released at <https://github.com/yianan261/IrresponsibleAI>; data visualization can be found at <https://observablehq.com/@irresponsible-ai/outline-for-ai-incident-visualization-notebooks>

² <https://incidentdatabase.ai/>

primarily focuses on the development of a robust classification methodology to categorize a vast volume of technology-related incidents from AIID.

Our objective is to streamline the process of the incidents classification for data visualization. However, the sheer volume of incidents—exceeding 600 cases—renders manual classification impractical. To address this challenge, we propose leveraging large language models (LLMs) to automate the classification process.

Pre-trained transformer language models, such as GPT-3 and GPT-4, demonstrate great in-context learning capabilities through natural language task description [2]. Those LLMs exhibit breakthrough performances in various downstream tasks such as text-interpretation through semantic processing abilities [4] and medical diagnoses or theorem proving through its reasoning capabilities by general deduction [13]. We harness the power of these Artificial Intelligence Generated Content (AIGC) tools for our classification task by providing the data from AIID to the LLM and use various prompting techniques to refine the results. In our use-case, it is critical to ensure that the automated classifications are not only well-supported by the data given, but also closely aligned with human judgment and standards. This necessitates a validation process to ascertain the reliability and trustworthiness of the classification outcomes against established human benchmarks.

By utilizing LLM technology, we aim to develop a scalable and efficient approach to classify these AI-related incidents. Our efforts are geared towards providing insights into the prevalence, trends, and implications of AI-related harm, thereby contributing to the advancement of responsible AI researches.

AI engineering is an emerging field, and consequently, AI incidents documentation practices are still evolving. Our contributions to the field include:

- An overview of three prompting techniques we tested for classifying AI-related incidents by a predefined taxonomy of classes of harm, geographic location, societal impact, intended application, field of deployment, and others. We evaluate the three different techniques and the results and compare the LLM classifications to our human manual classifications.
- A classification of 17 incidents from AIID by LLM using various prompting techniques and another benchmark classification set by human manual efforts. Future efforts will aim to apply the most appropriate classification technique to the full 628 incidents on AIID and use those results for data visualization as part of the Responsible AI study.
- Evaluation and validation of the classification results by two domain experts, which assess the capability of the LLM in this classification task.

2 Related Work

2.1 AI Incidents Documentation

The documentation of AI failures has become increasingly crucial as AI technologies are more widely deployed in sensitive and critical contexts. To improve the

quality of AI systems and prevent future incidents, the AI engineering community needs a comprehensive record of previous AI failures[10]. Various databases have been established to document AI incidents.

- **The AI and Algorithmic Incident and Controversies (AIAAIC) Repository** An independent initiative that logs global AI incidents and controversies, managed by Charlie Pownall, and is used by over 60 universities and organizations ³.
- **The AI Vulnerability Database (AVID)** The database focuses on cataloging proven high-level failure modes and specific vulnerabilities, providing valuable insights for AI engineers and auditors ⁴.
- **Awful AI** A curated list to track current harmful usage of AI, aiming to raise awareness about the dark side of AI technologies ⁵.
- **AI Incidents Database (AIID)** The most utilized database for reporting AI incidents ⁶. It encompasses a wide range of AI-related harms affecting physical health, safety, and social/political systems. This database also serves as the primary resource for our study.

Research using the records, including efforts to categorize the causes of AI failures [12] and a sector-based approach analysis of incidents [3], underscores the need for systematic approaches to classify and analyze these events. Despite the availability of databases cataloging numerous AI failures, similar incidents continue to occur. For instance, there are recurring instances where facial recognition systems disproportionately misidentify people of color, leading to wrongful accusations and arrests, despite widespread documentation and acknowledgment of bias in AI systems. Industry practitioners face significant challenges in effectively preventing and mitigating bias, often only identifying these serious issues after deployment [14].

These studies indicate that there is still much to learn and significant work to be done to develop a robust classification model capable of handling the complexity and variety of incident reports.

2.2 Large Language Models

With the advent of *pre-trained* language models (LM) with fixed architecture came a new paradigm shift in the learning of these models. LMs built on transformer architecture no longer rely on objective engineering nor fine-tuning to perform specific downstream tasks [9]. In the new paradigm, pre-trained LMs can solve various tasks with only textual *prompts* [9]. Previous works that have shown how LLMs can adapt and recognize tasks at inference time, also referred to as *in-context* learning [2]. These large models showed improved ability to learn a task from contextual information, where a model is guided by textual cues to

³ <https://www.aiaaic.org/aiaaic-repository>

⁴ <https://avidml.org/>

⁵ <https://github.com/daviddao/awful-ai>

⁶ <https://github.com/responsible-ai-collaborative/aiid>

perform a task. By processing an instruction or a handful of task examples, the model then predicts subsequent actions within the same context.

Specifically, in the domain of classification, LLMs have been applied with some success. For example, recent studies have deployed LLMs for classifying legal documents using detailed criteria [15] and have evaluated the GPT family of models for their effectiveness in biomedical reasoning and classification tasks [6]. The flexibility of LLMs to parse and organize complex data has facilitated their adoption in various burgeoning fields. However, relying entirely on LLM responses can pose risks, such as generating misleading answers in scenarios like academic forums, where models may produce incorrect responses if they fail to understand a question [18]. To mitigate these risks, type-specific prompting has been proposed to enhance the performance of GPT models in classification tasks. Our research builds upon this foundation by utilizing LLMs to automate the classification of AI incidents according to a predefined taxonomy, employing diverse prompting techniques to enhance this process.

3 Background

Previously introduced in Section 2.1, the dataset used in our study is sourced from the AI Incidents Database (AIID) GitHub repository. This dataset was chosen due to its extensive collection of incidents, featuring over 1,000 archived reports from more than 600 contributors across various sectors. These incidents document real-world cases where the use of AI has resulted in harm, covering a range of issues from facial recognition and targeted advertising to collisions involving autonomous vehicles, among many others.

Before introducing our task setup, we present the main classification fields that our LLM aims to identify. The predefined taxonomy for this study encompasses classes of AI-related harm ranging from discrimination, disinformation, human incompetence, pseudoscience, copyright violation, mental health implications, to environmental impact. In addition to the taxonomy of classes of harm, the LLM should also identify other fields of classification germane to our study of Responsible AI.

3.1 Classification Fields Explained

To clearly define the scope and detail of the data the LLM analyzes in each incident, we outline the specific fields it is designed to classify:

- **Geographical Location (Country, State, City, Continent):** These fields determine an incident’s location, with the possibility of indicating "Worldwide" for events not confined to a single geographical area.
- **Company and Location (Company, Company City, Company State):** This identifies the organization responsible for the AI technology involved in the incident, including the precise location of its headquarters.

- **Affected Population:** This field aims to identify which demographic groups were impacted by the incident, shedding light on the societal segments most at risk from irresponsible AI usages.
- **Number of People Affected:** Both the actual and potential number of individuals impacted by the incident are quantified to gauge its severity and reach.
- **Taxonomic Classification of Classes of Harm (Classes, Subclasses, Sub-subclass):** Through a structured taxonomy, incidents are categorized into specific classes and subclasses of AI-related harm, enabling a systematic analysis of the types of irresponsible AI usage occurring. This taxonomy was collectively decided by the team and serve as labels or tags for the incident classification.
- **Application Area:** The specific domain of AI application involved in the incident is identified, such as healthcare, surveillance, or social media, providing insights into where AI poses the greatest risks.
- **Online:** Indicates whether the incident took place online, highlighting the digital nature of many AI-related ethical concerns.

Each classification field plays a crucial role in our comprehensive analysis of AI-related incidents from AIID, facilitating a nuanced understanding of how and where AI technologies are causing harm.

3.2 Taxonomy

We adopted the comprehensive taxonomy developed by Dr. Ricardo Baeza-Yates as the classification framework for analyzing our dataset of incidents (see Figure 1). The taxonomy outlines the various classes of harm resulting from irresponsible AI use, forming a hierarchical structure where the main categories represent the broad types of harm, and the sub-classes offer a more detailed breakdown of these categories. For instance, 'Discrimination' serves as a main category that can be further dissected into 'Data Bias' or 'Algorithmic Bias.' Within 'Data Bias,' specifics such as gender, race, sexual orientation, economics, and others are identified as sub-subclass. This taxonomy will serve as the labels for the classes of harm in our classification.

Automation with Large Language Models (LLMs): We programmed the LLMs to recognize and classify incidents according to the taxonomy, leveraging their natural language capabilities to assess the incidents' attributes and categorize them accordingly.

Human Classification: Simultaneously, a manual classification process was conducted. This involved a thorough review of each incident by the members of the team who then classified them based on the detailed criteria set out in the taxonomy. This process was not only used to validate the automated classifications but also ensured that nuanced or ambiguous cases were properly categorized.

Irresponsible AI Taxonomy

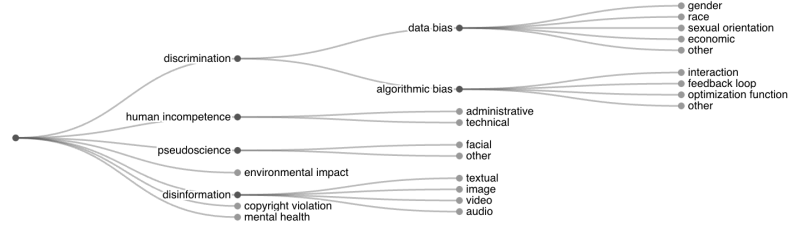


Fig. 1. Taxonomy for Classifying the Impacts of Irresponsible AI Utilization

4 Curation Methodologies

4.1 Experiment Setup

GPT-4 Model For our project, we utilized the Chat Completions API ⁷ by OpenAI to leverage the capabilities of an advanced language model for classifying the AI-related incidents dataset. The model of choice for our classification task was the GPT-4 model [11]. The GPT-4 language model represents an advancement over its predecessor, GPT-3, featuring significantly more parameters. GPT-4’s underlying technology is based on transformer architecture—specifically, attention mechanisms that emphasize different parts of the input data depending on the context. This model has been further refined with a technique called reinforcement learning from human feedback (RLHF), where human judgments help to fine-tune the model’s responses [5]. Although we specifically selected the GPT-4 model for this study, our future work may involve working with different models and evaluating their performances comprehensively.

Input Data From our dataset, we concentrated on eliminating redundant entries from the dataset. We merged all URLs linked to a specific incident under a singular incident ID, as the same incident might be reported by multiple news sources. This de-duplication process significantly streamlined the dataset, ensuring that each unique incident was catalogued as a single, exhaustive entry.

The result is a curated dataset devoid of redundancies, comprising 628 incidents⁸. Each incident is distinctly identified by an incident ID, consolidating all pertinent URLs under this unique identifier. This enables the classification to be based on the substantive content of the articles associated with each incident.

Next, we aggregated all available URL links, drawing reports from various news sources, social media posts, or official statement, for each incident within

⁷ <https://platform.openai.com/docs/api-reference/chat/create>

⁸ <https://observablehq.com/@irresponsible-ai/aiid-data-processing>

the AIID database. Each URL serves as a crucial source of information for specific incidents, offering additional context, details, and diverse perspective. Using Newspaper3k article scraping library⁹, we fetched the HTML content of each web-page, parsed it to extract relevant article text, and filtered out noise or irrelevant content. The extracted text was concatenated into a single comprehensive string for each incident, which then fed into the LLM as the input. By providing a substantial corpus of text data from multiple sources, the model is able to gain exposure to diverse language patterns, enhancing its ability to accurately understand and classify incidents.

4.2 GPT-4 Configurations

A critical parameter in our use of the Chat Completions API was the setting of the *temperature*, which influences the model’s output variability. The temperature parameter can range from 0 to 1, with lower values producing more deterministic and less varied outputs, and higher values encouraging more creativity and diversity in the generated text¹⁰. For the scope of our study, which centered around the straightforward tasks of text interpretation and classification, we adjusted the temperature setting from 0.5 to 0.3. This adjustment was made with the intention of minimizing the model’s propensity for generating creative responses, thereby ensuring that the output remained closely aligned with the factual content of the articles and the classification taxonomy.

4.3 Prompting Strategies

Delimiters The first few prompts did not have clear delimiters so sometimes the model was unable to understand the prompt instructions. Later we used delimiters to clearly indicate distinct parts of inputs¹¹. This approach significantly enhanced the model’s ability to comprehend and respond to the prompts as intended.

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1 ===== Start of Article Content =====
2               {article_text}
3 ===== End of Article Content =====

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Listing 1.1. Using Delimiters in Prompts

Persona-Based Prompting To guide the behavior of the language model, we employed a strategy of persona adoption, where the model is instructed to assume a specific role within its operational framework. For instance, we asked the model to operate as "a helpful assistant designed to classify news articles into

⁹ <https://newspaper.readthedocs.io/en/latest/>

¹⁰ <https://platform.openai.com/docs/guides/text-generation/how-should-i-set-the-temperature-parameter>

¹¹ <https://platform.openai.com/docs/guides/prompt-engineering/tactic-use-delimiters-to-clearly-indicate-distinct-parts-of-the-input>

specific categories," with the expected output format being JSON. This approach is designed to orient the model towards a particular behavior pattern¹²

Role Prompting for Enhanced Performance : The technique of role prompting leverages on the known sensitivity of Large Language Models (LLMs) to the construction of prompts. LLMs' performance can significantly benefit from the specification of roles within prompts, improving by at least 20 percent over control prompts where no specific context is provided [20]. This sensitivity underscores the effectiveness of role prompting in enhancing the model's task alignment and output accuracy.

We assigned an occupational role to our LLM. The initial prompt encouraged the LLM to conceptualize itself as part of a computer research institute tasked with categorizing incidents related to the irresponsible use of AI technology. Subsequently, we refined the prompting strategy by directly addressing our LLM as a computer researcher engaged in the same categorization task, eliminating the use of "imagine." This progression in prompt specificity was informed by evidence suggesting that LLMs respond with greater precision to direct role assignment, a reflection of their sensitivity to the nuances of prompt phrasing [20]. This iterative refinement in role prompting tailors the model's focus and improves its classification performance.

Detailed Step Instructions In the refined prompts, we detailed the steps necessary for task completion by specifying the actions required to achieve a desired outcome¹³. This modification improved GPT-4's compliance with the given instructions, a significant enhancement over earlier versions of the prompts that lacked such explicit direction. Figure 2 shows the prompt after clear steps are outlined. Previously, the model occasionally deviated from the prescribed format or introduced its own categories, despite explicit instructions to adhere to the predefined taxonomy. The incorporation of clearly defined steps and rules in the prompts resulted in outputs that more accurately reflected the intended instructions, demonstrating the LLM's increased ability to follow directions precisely.¹⁴

By adding step-by-step instructions refined the model's performance, the model was able to adhere to our instructions more closely and maintained a more consistent output format, whereas in the former zero-shot prompt some of the output format deviated from our specifications.

4.4 Prompting Techniques

Zero-Shot and Few-Shot Prompting The work of Kojima et al. [8] has shown that these LLMs possess remarkable capabilities for reasoning without

¹² <https://platform.openai.com/docs/guides/text-generation/chat-completions-api>

¹³ <https://platform.openai.com/docs/guides/prompt-engineering/tactic-specify-the-steps-required-to-complete-a-task>

¹⁴ See appendix ??

Prompting without steps	Prompting with added steps
Imagine a computer research institute trying to categorize a list of incidents of irresponsible use of artificial intelligence technology. Given the aggregated news article texts on relevant incidents, please extract the following information, your responses should be well thought-out and well-supported by the content of the articles, please also follow instructions of this prompt. Refer to {exampleOutput} for json output examples.	1. Take a look at this taxonomy and take a close look at these examples 2. Your task is mainly this part. You have to fill out the following fields according to the article content using the three steps above: STEP 1: Read the article text: STEP 2: State your reason for your classifications for the following: STEP 3: Make sure your classification output follows such format:

Fig. 2. Example of prompting without steps and prompting with added steps

prior exposure to specific data (zero-shot reasoning) and for rapidly adapting to tasks when provided with a small number of targeted examples (few-shot learning). The complexity of the task and capability of the model could influence how useful it is to include some examples in the prompt, as this may offer extra context and direction, potentially improving model performance [5]. However, it has also been shown that a well-designed zero-shot prompt, such as adding an intermediate reasoning step [8], could even lead to better performance and outperform few-shot prompts [5]. In our experiment, the few-shot learning with examples enabled GPT-4 to yield results that aligned more closely to the manual classifications¹⁵.

Chain of Thought Chain-of-Thought (CoT) prompting is introduced in the work of Wei et al. [16] that by eliciting an intermediate natural language reasoning step, the prompt can expand the set of diverse tasks that LLMs can perform successfully. Other research literature have shown that LLMs, combined with in-context learning (ICL) and chain-of-thought (CoT) prompting, are capable of various high-level reasoning tasks. [8] also introduced by simply adding "Let's think step by step" to a zero-shot prompt, the model outperforms standard zero-shot prompting. In our classification task, we set out with standard zero-shot prompting and also zero-shot prompting combined with Chain-of-Thought to elicit reasoning steps. We find that in the case of zero-shot prompting, there was not a big difference in results whether or not we added the reasoning steps. How-

¹⁵ See appendix ??

ever, adding reasoning steps in few-shot learning prompts significantly improved the classification outcomes.

Tree of Thoughts The Tree of Thoughts (ToT) concept was introduced in Yao et al. [17], which enables LLMs to perform deliberate problem-solving by maintaining and exploring a tree of thought steps. ToT frames problem-solving as a search over a tree of coherent language sequences ("thoughts") that serve as intermediate steps toward the solution, allowing exploration of multiple reasoning paths. This process bears resemblance to the thought and reasoning process typical to humans in that it encourages the LLM to plan and even backtrack in its reasoning to assess different paths of thoughts. It has been demonstrated that the ToT framework significantly outperforms standard input-output (IO) prompting and Chain-of-Thought prompting on certain reasoning and creative writing tasks [17].

Our classification task’s prompting methodology draws inspiration from these research insights. We present our observations from employing various prompting strategies with modifications and ensemble approaches that combine select techniques.

5 Output

5.1 Zero-Shot Learning

In the zero-shot learning results, GPT-4 demonstrated proficiency in pinpointing geographic and location information from the articles with considerable accuracy. However, the model occasionally misidentified the city associated with a company if it was unable to extract the location information from the text. The "Affected population" category displayed a broader range of responses and was notably more descriptive in describing the impacted group of people than the few-shot learning results. Additionally, GPT-4 was able to identify the hierarchy of harm classes along with their subcategories fairly reasonably. Nevertheless, its categorization at times varied from those done manually ¹⁶.

5.2 Few-Shot Learning with Chain-of-Thought Technique

In our few-shot learning prompt, we included two examples in the prompt featuring article texts along with their example classification results. We also integrated the Chain of Thought (CoT) approach by providing a supporting rationale for each classification outcome. It has been shown that LLMs are capable of producing sequences of reasoning when provided with examples that showcase this process in the context of few-shot prompting [16]. The purpose of these examples

¹⁶ See appendix E for the full result on the 17 incidents and appendix F for table comparisons at <https://github.com/Liheng-Yi/Appendix-For-Advancing-AI-Incidents-Classification-Leveraging-LLMs/tree/main/Appendix>

is to guide the models in replicating a similar style of reasoning. By enumerating reasoning step examples, the model is guided to arrive at a particular answer by extracting information from the article. Eliciting reasoning steps from the model also allows it to correct its own answers [16]. Our findings reveal that GPT-4 was generally accurate in identifying geographic and location information, occasionally misidentifying company city when that information was not clear nor present in articles. GPT-4 also reliably classified "Affected population" and "Area of AI Application," which are categories that allow a wider range of answers without fixed labels. With reasoning guidance provided in the examples, GPT-4's identification of the classes of harm were more consistent with manual classifications than with the results from zero-shot learning ¹⁷.

5.3 Tree of Thought Framework with Few-Shot Learning and Chain-of-Thought Technique

In addition to zero-shot and few-shot learning with CoT reasoning, we used an ensemble approach that combined few-shot learning and CoT reasoning with the Tree of Thoughts (ToT) framework. In the prompt, we asked GPT-4 to act as three experts in order to branch out into three reasoning and classification paths. The prompt guides the model to create a breadth-first search of the tree of probable classifications and ask the experts to vote on the classification is the most promising and well-supported by the article text to use that as the final result. Reference [17] mentions the "self-reflection" approach, wherein language models can provide feedback on the outputs they generate. The ToT framework leverages this capability to get the model to evaluate and select among its generated reasoning and outcome branches. We found GPT-4 was generally successful in identifying geographic location, but there were instances where company cities were misidentified if company location was not clearly mentioned in the article. Figure 3 shows that location categories scored highest in accuracy when compared to manual classifications. The model also successfully identified the "Affected population" and "Area of AI Application" fields. The classifications for the "Affected Population" provided by the ToT method were generally more succinct and concise because the exemplars referenced were in such style, compared to the zero-shot learning classification. Interestingly, when integrating the Tree of Thoughts (ToT) with the chain-of-thought technique, the average classification outcomes across all fields scored higher in semantic similarity with the manual classifications than the aforementioned approaches. ¹⁸

¹⁷ See appendix C for full result on 17 incidents at <https://github.com/Liheng-Yi/Appendix-For-Advancing-AI-Incidents-Classification-Leveraging-LLMs/tree/main/Appendix>

¹⁸ See appendix D for full result on 17 incidents at <https://github.com/Liheng-Yi/Appendix-For-Advancing-AI-Incidents-Classification-Leveraging-LLMs/tree/main/Appendix>

6 Evaluation

Evaluation of the efficacy of prompt methods in LLMs are mainly divided to subjective and objective categories. Subjective evaluations generally rely on human evaluators to assess the quality of the generated content. Such evaluation methods are subjective and can be prone to inconsistencies [5]. In some cases, human evaluation is the better way to evaluate the accuracy and quality of LLM generated results because evaluators (typically experts) can assess the results more comprehensively with more accurate feedback [4]. On the other hand, objective evaluation, or automatic evaluation methods include the use of machine learning algorithms to score the quality of the text generated by the LLMs. However, these automated metrics sometimes fail to fairly capture the full semantic considerations and cultural context in generated text as well as assessment results of human evaluators.

6.1 Objective Evaluation

We measured the accuracy score of the classification results from LLM from the manual results (used as the true labels). This evaluation metric focuses strictly on the literal exactness of the LLM’s outputs, but it does not consider semantic meanings or contextual nuances. Consequently, only single-word fields such as "State", "Company State", "City", tend to score high with this metric. We find that the few-shot learning with ToT and CoT prompt results had the highest matches for these fields, as shown in figure 3. This suggests that few-shot learning with ToT and CoT has the closest reasoning process to our manual classifications. Other fields such as "Affected population", "Area of AI application", and the classes of harm are expected to have lower scores due to the literal nature of this accuracy measurement. In addition to accuracy, we used BERTScore [19], which measures semantic similarity between pairs of text sequences. It provides a more nuanced evaluation compared to taking the accuracy score or other traditional evaluation metrics that rely more on matching the exactness of n-grams. The F1 scores are relatively high across all fields for each prompting technique as seen in figure 4, suggesting that the BERT model is able to capture the similarity between the evaluated text and the reference. The few-shot with ToT and CoT results tends to have the highest or near-highest F1 score in almost every field compared to the other sources, which may suggest that this approach is better at aligning with the manual evaluation. Although BERTScore is better at capturing semantic nuances, it still does not take on full contextual and semantic meanings, which accounted for the lower F1 scores across the classification fields for some categories.

6.2 Subjective Evaluation

Since our classification task is domain-specific to the study of Irresponsible AI and its impact, we used human subjective evaluation from domain experts to assess the quality of GPT-4’s output. Building on the experimental framework

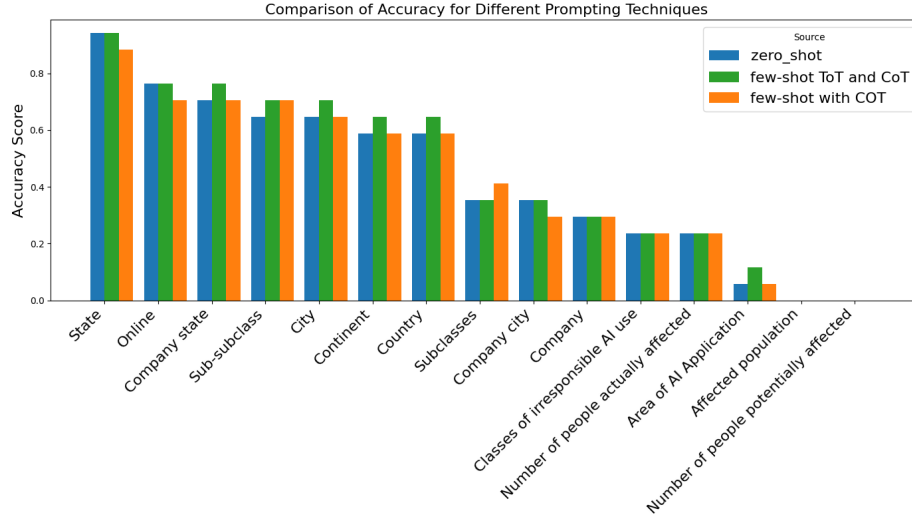


Fig. 3. Accuracy score of GPT-4 results. The accuracy scores here are measured by the exactness of literals using the manual classification results as the true label.

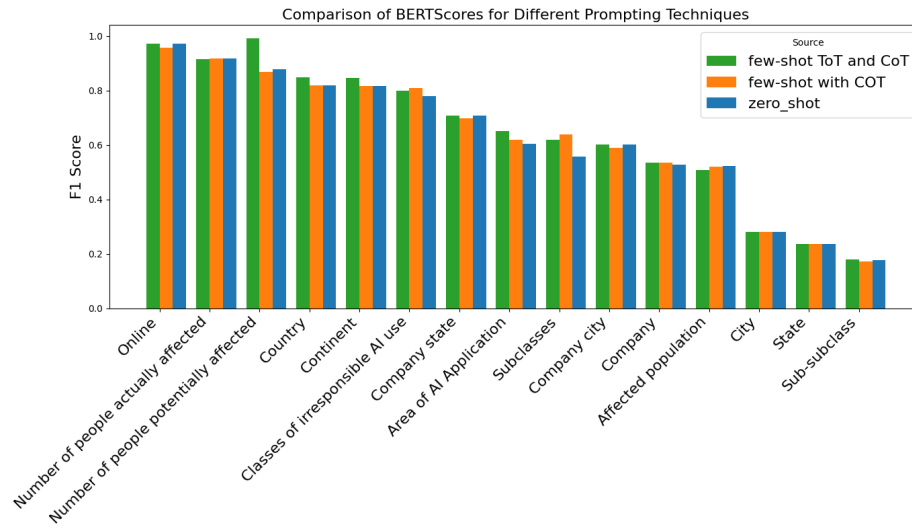


Fig. 4. F1 scores for different prompting technique results evaluated by the BERTScore evaluator

outlined in Section 4, our evaluation method involved a blind review by two independent domain experts. Each expert assessed the classifications—both human and LLM-generated—across various dimensions such as geographic location, affected population, incident impact, application of AI use, and the classes of harm. The reviewers were unaware of which incidents have been classified by GPT-4 and which fields were manually classified by the team. The classifications were scored on a scale from 1 to 10, based on accuracy of identification of each field. The average scores of the LLM classifications using the Tree-of-Thought ensemble approach prompting on 11 incidents achieved a slightly higher average score (at 7.73) than the average score of human manual classifications on 9 incidents (at 7.17). For both reviewers, the median scores for LLM classifications appear higher than for human classifications as can be seen in figures 5 and 6. This suggests that, on average, both reviewers rated LLM classifications more favorably than human ones. In both ratings, the variability of LLM classification ratings are generally wider than human classifications, indicating the reviewers both perceived some level of inconsistency in performance of the LLM results. Reviewer feedback on human classifications often cited issues such as typographical errors, incorrect identification of places, and lack of consensus regarding the labeling of harm classes, highlighting that human-generated results can also be prone to errors. On the other hand, the feedback for LLM classifications commonly mentioned the absence of predicted potential impacts, suggesting that the model may not have discerned certain information from the text or the information was not present for the model to make a judgment.

In some classification output, we have seen that GPT-4 is capable of producing very consistent output that is very close to human classification results ¹⁹. Other cases show that the results varied with different prompts. The outcome of the assessment suggests that LLMs can outperform human efforts in classification tasks in some cases, as shown by the higher average scores. However they are not without limitations, such as failing to predict certain outcomes from information gleaned from article texts. The variability and errors in human classifications also demonstrate the complexities and challenges that we often see in the manual categorization processes. This comparison suggests that while LLMs show promise as efficient tools for classification, combining their strengths with human oversight could offer a more robust approach.

It is worth noting that this experiment represents a small subset of the 628 incidents documented in the AIID, meaning the findings might not sufficiently reflect the broader performance capabilities of LLM and human classifications. We plan to experiment in small batches before we proceed with a suitable approach that could successfully classify all the incidents. Additionally, the variability in evaluations among expert reviewers might indicate the need for a more robust evaluation design, since it is inevitable for subjective evaluations to have wide variability because of differences in opinions.

¹⁹ See appendix ??

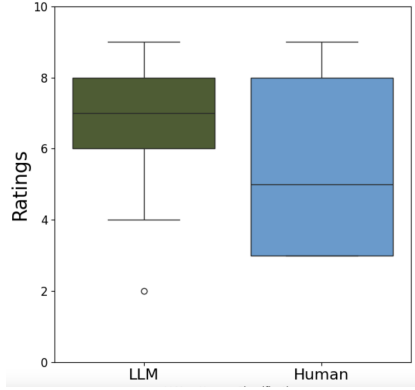


Fig. 5. Reviewer 1 ratings: distribution of reviewer 1's ratings on incidents classification from a blind review

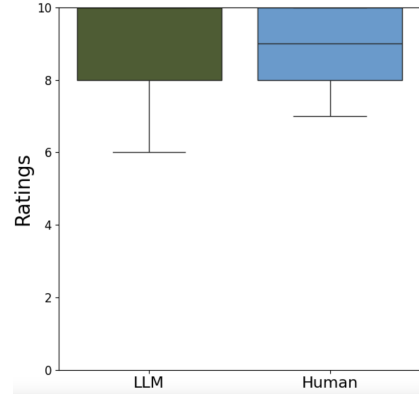


Fig. 6. Reviewer 2 ratings: distribution of reviewer 2's ratings on incidents classification from a blind review

7 Conclusions

Our study's focus was to enhance classification methods for the numerous AI incidents documented in the AI Incidents Database, aiming to improve understanding and management of AI-related harm.

The study leveraged the advanced capabilities of large language models (LLMs) like GPT-4 to classify incidents more efficiently than manual methods. Our findings indicated that while LLMs can sometimes outperform human classification in terms of average scoring, they are not infallible, occasionally misidentifying locations or missing potential impacts. This underscores the need for human oversight to ensure the highest quality and reliability in classification.

The investigation into prompt engineering, spanning zero-shot to few-shot learning with Chain of Thought and Tree of Thoughts frameworks, revealed nuances in model performance. Combining Tree of Thoughts method with few-shot prompts and explicit reasoning steps offered results most closely aligned with manual classifications. There are still known limitations to LLMs such as GPT-4. Most notably, like its predecessors, it still lacks complete reliability, sometimes producing fabricated information (known as "hallucinations") or committing logical errors [11]. Hallucinations occur because the model might not have enough supporting evidence in its training data for its responses, or it could generalize patterns too broadly to produce coherent output [7]. In some instances, we observed GPT-4 exhibiting this behavior where it could not accurately identify specific classification fields from the text of an article. For instance, it mistakenly identified the city of the company Chai Research²⁰ as "Silicon Valley" in one incident, when it is actually located in Palo Alto²¹.

²⁰ www.chai-research.com

²¹ See appendix ??

7.1 Future Work

Moving forward, we plan to extend our methodology to the entire corpus of incidents in AIID and apply a more robust and effective approach to classify all the news incidents with high accuracy, which will be corroborated by experts in the field. Our aim is to harness these classifications to create data visualizations that will contribute to the ongoing discourse on Responsible AI, providing greater insight into AI-related harm trends and influencing the development of more responsible AI systems.

This work serves as a step toward understanding and documenting AI failures and establishing a robust, automated classification framework to handle the complexity of AI incident reports. It highlights both the capabilities and limitations of LLMs in this domain and points to a future where collaboration between human expertise and AI could offer robust solutions for managing AI's societal impacts.

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References

1. Baeza-Yates, R.: LECTURE HELD AT THE ACADEMIA EUROPAEA BUILDING BRIDGES CONFERENCE 2022: An Introduction to Responsible AI. *European Review* **31**(4), 406–421 (8 2023). <https://doi.org/10.1017/S1062798723000145>, <https://www.cambridge.org/core/journals/european-review/article/lecture-held-at-the-academia-europaea-building-bridges-conference-2022/C61F82D2FF64FFB933282F5E65CDB22A>
2. Brown, T.B., Mann, B., Ryder, N., Subbiah, M., Kaplan, J., Dhariwal, P., Neelakantan, A., Shyam, P., Sastry, G., Askell, A., Agarwal, S., Herbert-Voss, A., Krueger, G., Henighan, T., Child, R., Ramesh, A., Ziegler, D.M., Wu, J., Winter, C., Hesse, C., Chen, M., Sigler, E., Litwin, M., Gray, S., Chess, B., Clark, J., Berner, C., Mccandlish, S., Radford, A., Sutskever, I., Openai, D.A.: *Language Models are Few-Shot Learners* (2020)
3. Burema, D., Debowski-Weimann, N., Von Janowski, A., Grabowski, J., Maftai, M., Jacobs, M., Van Der Smagt, P., Benbouzid, D.: A sector-based approach to AI ethics: Understanding ethical issues of AI-related incidents within their sectoral context. *AIES 2023 - Proceedings of the 2023 AAAI/ACM Conference on AI, Ethics, and Society* pp. 705–714 (8 2023). <https://doi.org/10.1145/3600211.3604680>, <https://dl.acm.org/doi/10.1145/3600211.3604680>
4. Chang, Y., Wang, X.U., Yi, X., Wang, Y., Ye, W., Yu, P.S., Chang, Y., Wang, X., Wu, Y., Yi, X., Xie, X., Yang, L., Wang, C., Zhang, Y., Wang, Y., Ye, W., Wang, X., Wang, J., Wu, Y., Yang, L., Zhu, K., Chen, H., Wang, C., Zhang, Y., Chang,

- Y., Yang, Q., Xie, X.: A Survey on Evaluation of Large Language Models. *ACM Trans. Intell. Syst. Technol* **15**(3), 39 (2024). <https://doi.org/10.1145/3641289>, <https://github.com/MLGroupJLU/LLM-eval-survey>
5. Chen, B., Zhang, Z., Langrené, N., Zhu, S.: Unleashing the potential of prompt engineering in Large Language Models: a comprehensive review (10 2023), <http://arxiv.org/abs/2310.14735>
 6. Chen, S., Li, Y., Lu, S., Van, H., Aerts, H.J.W.L., Savova, G.K., Bitterman, D.S.: Evaluating the ChatGPT family of models for biomedical reasoning and classification. *Journal of the American Medical Informatics Association* (4 2024). <https://doi.org/10.1093/JAMIA/OCAD256>
 7. Ji, Z., Lee, N., Frieske, R., Yu, T., Su, D., Xu, Y., Ishii, E., Bang, Y.J., Madotto, A., Fung, P.: Survey of Hallucination in Natural Language Generation. *ACM Computing Surveys* **55**(12) (3 2023). <https://doi.org/10.1145/3571730>, <https://dl.acm.org/doi/10.1145/3571730>
 8. Kojima, T., Gu, S.S., Reid, M., Matsuo, Y., Iwasawa, Y.: Large Language Models are Zero-Shot Reasoners (5 2022), <http://arxiv.org/abs/2205.11916>
 9. Liu, P., Yuan, W., Fu, J., Jiang, Z., Hayashi, H., Neubig, G.: Pre-train, Prompt, and Predict: A Systematic Survey of Prompting Methods in Natural Language Processing (2021)
 10. Mcgregor, S.: Preventing Repeated Real World AI Failures by Cataloging Incidents: The AI Incident Database (2021), www.aaai.org
 11. OpenAI, Achiam, J., Adler, S., Agarwal, S., Ahmad, L., Akkaya, I., Aleman, F.L., Almeida, D., Altenschmidt, J., Altman, S., Anadkat, S., Avila, R., Babuschkin, I., Balaji, S., Balcom, V., Baltescu, P., Bao, H., Bavarian, M., Belgum, J., Bello, I., Berdine, J., Bernadett-Shapiro, G., Berner, C., Bogdonoff, L., Boiko, O., Boyd, M., Brakman, A.L., Brockman, G., Brooks, T., Brundage, M., Button, K., Cai, T., Campbell, R., Cann, A., Carey, B., Carlson, C., Carmichael, R., Chan, B., Chang, C., Chantzis, F., Chen, D., Chen, S., Chen, R., Chen, J., Chen, M., Chess, B., Cho, C., Chu, C., Chung, H.W., Cummings, D., Currier, J., Dai, Y., Decareaux, C., Degry, T., Deutsch, N., Deville, D., Dhar, A., Dohan, D., Dowling, S., Dunning, S., Ecoffet, A., Eleti, A., Eloundou, T., Farhi, D., Fedus, L., Felix, N., Fishman, S.P., Forte, J., Fulford, I., Gao, L., Georges, E., Gibson, C., Goel, V., Gogineni, T., Goh, G., Gontijo-Lopes, R., Gordon, J., Grafstein, M., Gray, S., Greene, R., Gross, J., Gu, S.S., Guo, Y., Hallacy, C., Han, J., Harris, J., He, Y., Heaton, M., Heidecke, J., Hesse, C., Hickey, A., Hickey, W., Hoeschele, P., Houghton, B., Hsu, K., Hu, S., Hu, X., Huizinga, J., Jain, S., Jain, S., Jang, J., Jiang, A., Jiang, R., Jin, H., Jin, D., Jomoto, S., Jonn, B., Jun, H., Kaftan, T., Kaiser, , Kamali, A., Kanitscheider, I., Keskar, N.S., Khan, T., Kilpatrick, L., Kim, J.W., Kim, C., Kim, Y., Kirchner, J.H., Kiros, J., Knight, M., Kokotajlo, D., Kondraciuk, , Kondrich, A., Konstantinidis, A., Kopic, K., Krueger, G., Kuo, V., Lampe, M., Lan, I., Lee, T., Leike, J., Leung, J., Levy, D., Li, C.M., Lim, R., Lin, M., Lin, S., Litwin, M., Lopez, T., Lowe, R., Lue, P., Makanju, A., Malfacini, K., Manning, S., Markov, T., Markovski, Y., Martin, B., Mayer, K., Mayne, A., McGrew, B., McKinney, S.M., McLeavey, C., McMillan, P., McNeil, J., Medina, D., Mehta, A., Menick, J., Metz, L., Mishchenko, A., Mishkin, P., Monaco, V., Morikawa, E., Mossing, D., Mu, T., Murati, M., Murk, O., Mély, D., Nair, A., Nakano, R., Nayak, R., Neelakantan, A., Ngo, R., Noh, H., Ouyang, L., O’Keefe, C., Pachocki, J., Paino, A., Palermo, J., Pantuliano, A., Parascandolo, G., Parish, J., Parparita, E., Passos, A., Pavlov, M., Peng, A., Perelman, A., Peres, F.d.A.B., Petrov, M., Pinto, H.P.d.O., Michael, Pokorny, Pokrass, M., Pong, V.H., Powell, T., Power, A., Power, B., Proehl, E.,

- Puri, R., Radford, A., Rae, J., Ramesh, A., Raymond, C., Real, F., Rimbach, K., Ross, C., Rotsted, B., Roussez, H., Ryder, N., Saltarelli, M., Sanders, T., Santurkar, S., Sastry, G., Schmidt, H., Schnurr, D., Schulman, J., Selsam, D., Sheppard, K., Sherbakov, T., Shieh, J., Shoker, S., Shyam, P., Sidor, S., Sigler, E., Simens, M., Sitkin, J., Slama, K., Sohl, I., Sokolowsky, B., Song, Y., Staudacher, N., Such, F.P., Summers, N., Sutskever, I., Tang, J., Tezak, N., Thompson, M.B., Tillet, P., Tootoonchian, A., Tseng, E., Tuggle, P., Turley, N., Tworek, J., Uribe, J.F.C., Vallone, A., Vijayvergiya, A., Voss, C., Wainwright, C., Wang, J.J., Wang, A., Wang, B., Ward, J., Wei, J., Weinmann, C., Welihinda, A., Welinder, P., Weng, J., Weng, L., Wiethoff, M., Willner, D., Winter, C., Wolrich, S., Wong, H., Workman, L., Wu, S., Wu, J., Wu, M., Xiao, K., Xu, T., Yoo, S., Yu, K., Yuan, Q., Zaremba, W., Zellers, R., Zhang, C., Zhang, M., Zhao, S., Zheng, T., Zhuang, J., Zhuk, W., Zoph, B.: GPT-4 Technical Report (3 2023), <http://arxiv.org/abs/2303.08774>
12. Pittaras, N., McGregor, S.: A taxonomic system for failure cause analysis of open source AI incidents (11 2022), <https://arxiv.org/abs/2211.07280v1>
 13. Saparov, A., Pang, R.Y., Padmakumar, V., Joshi, N., Kazemi, S.M., Kim, N., He, H.: Testing the General Deductive Reasoning Capacity of Large Language Models Using OOD Examples (5 2023), <http://arxiv.org/abs/2305.15269>
 14. Turri, V., Dzombak, R.: Why We Need to Know More: Exploring the State of AI Incident Documentation Practices. AIES 2023 - Proceedings of the 2023 AAAI/ACM Conference on AI, Ethics, and Society pp. 576–583 (8 2023). <https://doi.org/10.1145/3600211.3604700>, <https://dl.acm.org/doi/10.1145/3600211.3604700>
 15. Wei, F., Keeling, R., Huber-Fliflet, N., Zhang, J., Dabrowski, A., Yang, J., Mao, Q., Qin, H.: Empirical Study of LLM Fine-Tuning for Text Classification in Legal Document Review. Proceedings - 2023 IEEE International Conference on Big Data, BigData 2023 pp. 2786–2792 (2023). <https://doi.org/10.1109/BIGDATA59044.2023.10386911>
 16. Wei, J., Wang, X., Schuurmans, D., Bosma, M., Ichter, B., Xia, F., Chi, E., Le, Q., Zhou, D.: Chain-of-Thought Prompting Elicits Reasoning in Large Language Models (1 2022), <http://arxiv.org/abs/2201.11903>
 17. Yao, S., Yu, D., Zhao, J., Shafran, I., Griffiths, T.L., Cao, Y., Narasimhan, K.: Tree of Thoughts: Deliberate Problem Solving with Large Language Models (5 2023), <http://arxiv.org/abs/2305.10601>
 18. Zhang, P., Jaipersaud, B., Ba, J., Petersen, A., Zhang, L., Zhang, M.R.: Classifying Course Discussion Board Questions using LLMs. Annual Conference on Innovation and Technology in Computer Science Education, ITiCSE **2**, 658 (6 2023). <https://doi.org/10.1145/3587103.3594202>
 19. Zhang, T., Kishore, V., Wu, F., Weinberger, K.Q., Artzi, Y.: BERTScore: Evaluating Text Generation with BERT (4 2019), <http://arxiv.org/abs/1904.09675>
 20. Zheng, M., Pei, J., Jurgens, D.: Is "A Helpful Assistant" the Best Role for Large Language Models? A Systematic Evaluation of Social Roles in System Prompts. [arxiv.orgM Zheng, J Pei, D JurgensarXiv preprint arXiv:2311.10054](https://arxiv.org/abs/2311.10054), 2023•[arxiv.org](https://arxiv.org/abs/2311.10054) (11 2023), <http://arxiv.org/abs/2311.10054>